

MCA (Two Year Programme)
SEMESTER -- I
CSL4036: Machine Learning
(General Elective-I)

Max Marks: 30

Time: 1 hour

- Q1. Using a suitable example, describe the key steps involved in a typical machine learning workflow. Explain how data preprocessing, feature engineering, and model refinement contribute to improving model performance. (7.5)
- Q2. Discuss the concept of a Confusion Matrix and its importance in evaluating machine learning models. Provide a detailed example with calculations to illustrate the use of a Confusion Matrix. (7.5)
- Q3. Describe gradient descent and its application in linear regression. Explain how overfitting and underfitting affect the performance of a linear regression model. (7.5)
- Q4. Explain the concept of support vector machines (SVMs) in classification. How does a non-linear SVM classifier differ from a linear one? (7.5)